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Enhancing Drilling Efficiency: Real-Time Rheology and Density Profiling with RheoProfiler in Saudi Arabia's Ghawar Gas Field

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Abstract

As drilling operations evolve toward greater digitalization, the demand for real-time data, automation, and integrated workflows has significantly increased. This paper presents the large-scale deployment and operational integration of the RheoProfiler 200, an advanced, automated rheometer, in the Ghawar Gas Field—one of the world's most prolific and technically complex hydrocarbon developments. The RheoProfiler 200, in combination with the Drilling Fluids Advisor and a mature Real-Time Operations Center (RTOC) framework, was implemented to overcome long-standing challenges in drilling fluid monitoring, including delayed data, human error, limited data accessibility, and reactive decision-making.

Weighing approximately 34 kg and designed for use in rig-site cabins, the RheoProfiler 200 performs fully automated mud testing, measuring key parameters such as density, plastic viscosity (PV), yield point (YP), and gel strengths at 10 seconds and 10 minutes. It transmits data through WITSML protocols to centralized platforms, where the Drilling Fluids Advisor compares results in real time against planned limits established in the Digital Drilling Program. The system's intuitive interface allows 24/7 access to stakeholders, and its outputs feed directly into hydraulic modeling engines, enabling the calculation of Hole Cleaning Index, theoretical pump pressure, and Equivalent Circulating Density (ECD) across the entire wellbore.

The implementation was conducted across 11 rigs within one year, covering 48 wells, over 300 sections, and more than 750,000 ft of drilled footage. More than 15,000 automated rheological tests were performed during this period. The system proved critical in mitigating risks under complex conditions such as high-density fluid use, differential sticking, fluid losses, and high-pressure, high-temperature (HPHT) environments. Real-time fluid property control enabled proactive responses to surge/swab events during tripping and casing runs, while improved ECD monitoring reduced the need for aggressive mud dilutions.

One of the most significant advancements offered by this integration was the ability to move away from traditional, manual mud testing and printed 24-hour reports. Manual testing introduces delays, data silos, inconsistency, and limits the ability to react swiftly to rapid downhole changes. By contrast, the RheoProfiler 200 increased test frequency from 2–3 per day to up to 6 automated tests, with a data latency of under 2

minutes compared to the conventional 24-hour delay. Accuracy was confirmed through cross-validation, with deviations maintained within ±2% for density and ±5% for rheology.

The Drilling Fluids Advisor platform aggregates all planned and real-time fluid data, highlighting deviations and generating alerts using a standardized, intuitive "blood test" format. This format allows non-specialists—such as the Company Representative or remote engineers—to quickly grasp deviations and severity. Alerts are automatically distributed to key stakeholders, including field engineers, RTOC personnel, and Subject Matter Experts (SMEs), facilitating collaborative problem-solving and timely corrective action. More than 45 documented remote interventions occurred as a result of real-time anomaly alerts, demonstrating the system's ability to unlock centralized expertise in support of field operations.

Operationally, the integration of these systems enabled the prevention of eight high-density differential sticking incidents and the mitigation of seven well control events, all of which could have resulted in costly Non-Productive Time (NPT). The ability to precisely manage mud properties and ECD led to improved wellbore stability and reduced loss incidents during critical operations such as liner running in fragile formations. The project also delivered significant efficiency gains, with a reduction in manual mud testing time from 90 minutes per day to 15 minutes and estimated monthly savings of 60 man-hours per rig.

The digital maturity of the Ghawar project—including established SOPs, centralized digital infrastructure, and integrated applications—was pivotal in the seamless implementation of RheoProfiler and its associated systems. The project highlights how real-time fluid analytics, combined with automation and centralized collaboration, can enhance drilling performance, and establish a new benchmark for mud management in high-performance drilling environments.

This paper offers detailed insights into the technical configuration, field deployment, integration with digital platforms, and measurable impact of the RheoProfiler system. It provides a replicable model for the industry to adopt real-time fluid intelligence as a critical enabler of safe, efficient, and digitally transformed drilling operations.

Rheoprofiler

The RheoProfiler 200 is an advanced, fully automated rheometer specifically engineered for testing the density and rheological properties of drilling fluids, including water-based, oil-based, and synthetic mud systems. Weighing approximately 34 kg, the unit is designed for indoor operation and is ideally suited for installation at Fluids Engineer cabin at the rig site.

Figure 1—RheoProfiler Device in the lab test with the sample fluid container and touch screen.

This device plays a critical role in modern drilling operations by enabling consistent and reliable mud testing while minimizing manual intervention. It performs key measurements—including mud weight (density), plastic viscosity (PV), yield point (YP), and gel strengths (10-second and 10-minute)—and transmits the results via WITSML protocol to centralized data platforms such as Digital Platform. Although real-time data transmission is supported, results are typically reported with a short delay of a few minutes, depending on operational workflows.

OBM & WBM Checks	Density	Plastic Viscosity	Yield Point	Gels	Funnel Viscosity	F/L	Retort	Chemicals	PH	HTHP	PPT	Electrical Stability
RheoProfiler 200	Yes	Yes	Yes	Yes	No	No	No	No	No	No	No	No

Figure 2—Readings from RheoProfiler from the standard fluid test in the rig site.

Key Features

- User-Friendly Interface: A built-in touchscreen allows mud engineers to monitor data and make fluid treatment decisions quickly and efficiently.
- Automated Testing Capabilities: The system conducts unattended, repeatable measurements from multiple sampling locations, improving both consistency and laboratory throughput.
- Integrated Data Management: Each test is time-stamped and automatically uploaded to electronic drilling recorders and shared with Drilling Fluids Advisors for real-time oversight.

Benefits

- Comprehensive Mud Type Compatibility: Capable of analyzing all standard drilling fluids, the RheoProfiler 200 reduces the need for repetitive manual tasks and enhances safety.
- Precision Thermal Control: Equipped with advanced heating and cooling systems, the device ensures that fluid properties are measured under standardized temperature conditions, improving the accuracy and reliability of test results.
- Operational Efficiency: By automating routine testing, the device enables engineers to focus on higher-value tasks, supporting better decision-making and improving well construction performance.

RheoProfiler 200: Advantages Over Manual Testing and Traditional 24-Hour Mud Reporting

Manual mud testing and daily printed reports have long been the industry standard, but this approach presents a number of operational limitations that can negatively impact drilling performance and decision-making. The RheoProfiler addresses these issues with a fully automated, real-time fluid monitoring solution.

Challenges of the Outdated Manual Process

- Lack of Real-Time Data: With fluid measurements taken only a few times per day and reported with significant delay, there is minimal visibility into rapid changes during critical operations such as tripping, casing runs, or pressure transitions.
- Inaccuracy and Inconsistency: Manual testing is inherently prone to human error and subjectivity. Analog instruments can lead to small misreadings that remain undetected until they manifest as fluid-related problems downhole.
- Limited Data Integration: Manually recorded results are often siloed in printed reports or spreadsheets and are not easily integrated into rig automation systems, real-time operation centers (RTOCs), or digital analytics platforms.

- Delayed Decision-Making: The lag between sampling, testing, and reporting restricts the ability to respond proactively to fluid property changes, increasing the likelihood of drilling dysfunctions such as poor hole cleaning or wellbore instability.
- High Resource Demand: Manual testing is time-consuming and labor-intensive, occupying rig personnel with repetitive tasks and diverting focus from more strategic engineering responsibilities.

Project Selection to Implement Rheoprofiler

The Ghawar Gas Field, located in Saudi Arabia, is one of the world's largest and most productive conventional gas fields. Over the past 12 years, the field has been actively developed using 11 drilling rigs, with more than 300 wells drilled, accumulating over 5.5 million feet of footage. This extensive drilling campaign is supported by a mature Real-Time Operations Center (RTOC), which operates under well-defined standard operating procedures (SOPs) and robust communication protocols. The RTOC acts as a centralized hub for monitoring, analysis, and decision-making, enabling optimized field development and operational efficiency.

Throughout the course of this long-term project, the Ghawar Gas Field has undergone a significant digital transformation, aligned with the broader industry shift toward digital oilfield technologies. This transformation includes the implementation of Digital Drilling Programs, and a suite of fit-for-purpose digital applications tailored to support various roles across the drilling organization. These tools facilitate seamless collaboration among rig personnel, remote operations centers, and engineering teams, ensuring that critical data is accessible in real time and that decisions are evidence-based.

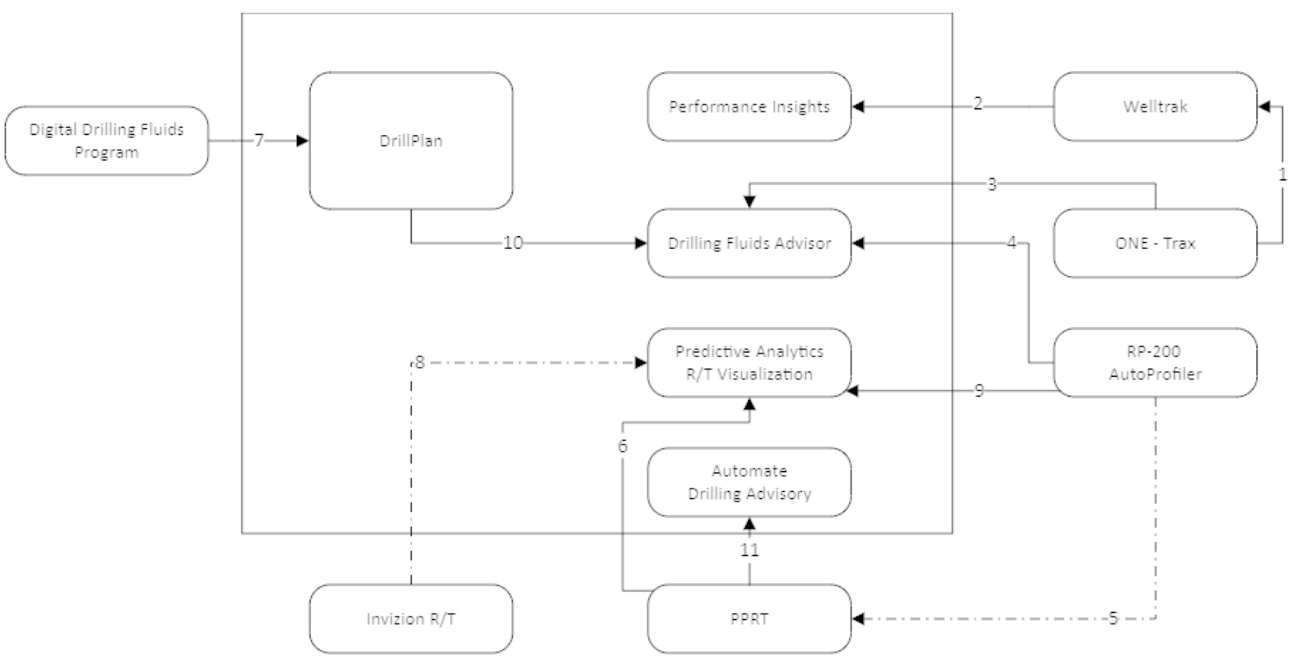

Figure 3—Robust Digital Platform with synchronized applications customized for the various stake holders

One of the critical challenges faced during drilling operations in the Ghawar fields has been managing unpredictable total fluid losses, and also in other sections drill with high-density muds and maintain significant overbalance pressure to control formation pressures. These conditions increase the risk of differential sticking, a common cause of stuck pipe incidents that can lead to costly non-productive time (NPT). Therefore, accurate and timely control of drilling fluid properties is essential to mitigate such risks and maintain drilling performance.

In response to these challenges, the RheoProfiler tool was introduced as part of the digital transformation initiative to improve drilling fluid management. The device measures key mud properties with high precision and consistency.

What sets the RheoProfiler apart is its ability to continuously transmit these measurements in real time to centralized monitoring platforms. This constant data stream allows for live fluid property analysis, early detection of trends, and immediate alerts to potential issues. Remote engineering and fluid support teams can then provide proactive recommendations, enabling timely interventions before problems escalate.

The availability of real-time fluid data drastically improves the decision-making process, allowing drilling engineers to adjust mud formulations and drilling parameters quickly. This capability reduces the response time to downhole changes during dynamic operations such as tripping, casing running, or pressure transitions, where fluid properties can fluctuate rapidly.

Moreover, by automating routine testing, the RheoProfiler frees rig personnel from repetitive manual measurements, reducing human error and increasing operational efficiency. The integration of RheoProfiler data into the digital workflow complements the existing RTOC capabilities, contributing to safer and more efficient well construction.

Integration of the RheProfiler into a robust Digital Platform

The presence of a robust and digitally mature project environment—with a Real-Time Operations Center (RTOC) supporting drilling activities for over a decade—greatly facilitated the seamless integration of the RheoProfiler system into existing data workflows and stakeholder applications. Full deployment of the RheoProfiler across all eleven operational rigs was accomplished within approximately one year.

During this implementation phase, the Drilling Fluids Engineer played a critical role in designing the fluid system for each well, considering formation characteristics, anticipated rate of penetration (ROP), hole size, and hole-cleaning efficiency. Based on this analysis, the engineer established operational boundaries and performance targets for key fluid properties. These included fluid density, yield point, rheological measurements (notably R_6 and R_3 readings), gel strengths at 10 seconds and 10 minutes, and the selection of the appropriate fluid system for each hole section.

These parameters are then delivered to the Design Well Engineer, who consolidates and verifies the data for completeness and accuracy. The Design Engineer subsequently generates the Digital Drilling Program, which is uploaded to the digital platform. This program serves as a foundational reference for all downstream applications, including the Drilling Fluids Advisor.

The Drilling Fluids Advisor aggregates all relevant fluid-related data from the Digital Drilling Program, with particular emphasis on predefined operational limits for key drilling fluid parameters. During execution, the Advisor continuously compares these planned limits with real-time measurements obtained from the RheoProfiler system. Additionally, the Advisor is fully integrated with the rig's reporting system, enabling it to cross-reference actual fluid properties reported in both the RheoProfiler and the daily reporting system against the planned values. This synchronization ensures enhanced monitoring, real-time validation, and proactive management of drilling fluid performance.

In addition to real-time rheological monitoring, the RheoProfiler was integrated with a Hydraulics Engine, enabling the generation of advanced deliverables such as the Hole Cleaning Index, theoretical pump pressure, and Equivalent Circulating Density (ECD) at any point along the wellbore. This functionality proved especially valuable during tripping operations, as it allowed the Real-Time Operations Center (RTOC) to recommend optimal tripping speeds, thereby minimizing the risk of surge or swab events.

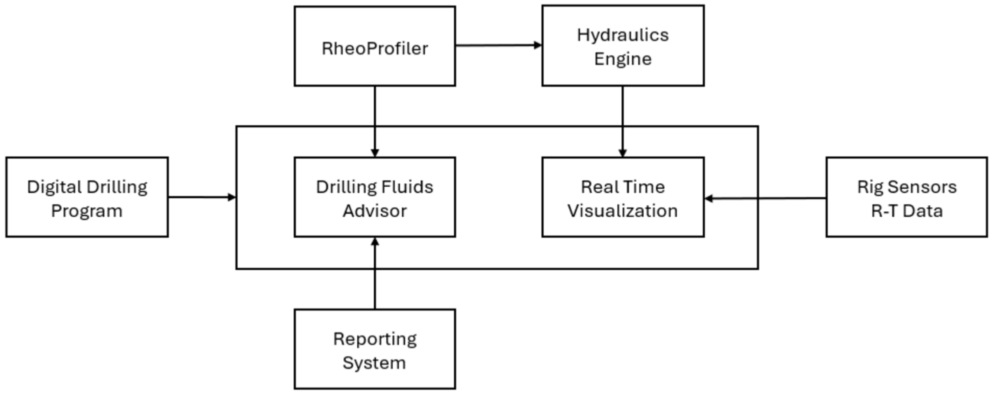

Figure 4—RheoProfiler Integration into the Real Time Digital Ecosystem

The Drilling Fluids Advisor platform is primarily utilized by the Execution Fluids Engineer; however, its data is accessible 24/7 to all stakeholders through the digital platform. Its user-friendly visualization tools empower wellsite and RTOC engineers to identify deviations in fluid properties efficiently.

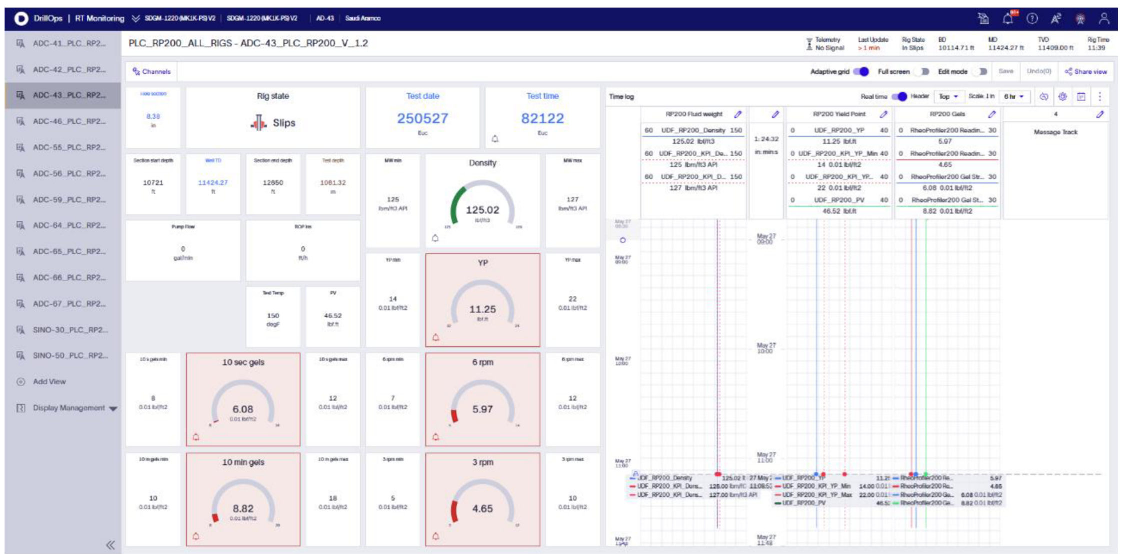

Figure 5—Digital Drilling Program limits are compared in real time with the RheProfiler real time data.

When deviations from planned fluid parameters are detected, the system automatically generates a standardized report and disseminates it to relevant stakeholders. Designed in a format analogous to a medical blood test, the report presents deviations in a clear and intuitive manner, enabling even non-specialists to quickly interpret the results and assess the severity of the anomaly.

PLC - WCF - Rheoprofiler 200 - Intervention Date: 27-May-25

Rig name: ADC-43L Fluid name: Rheliant
Well name : Test time: 08:21 AM
Hole section: 8.375

Property	Test result			Property specs		Observation
-	Below	Within	Above	Min	Max	-
Rheology T [°F]				150		
Fluid density [lb/ft³]		125.02		125	127	Within range
YP	11.25			14	22	Below range
R6	5.97			7	12	Below range
R3	4.65			5	10	Below range
10 s gels	6.08			8	12	Below range
10 min gels	8.82			10	18	Below range

Intervention criticality -Low

Properties within range	1
Properties exceeding performance indicators	0
Properties below performance indicators	5

Figure 6—Deviation Report in format analogous to medical blood test for easy understanding

The initial alert is sent to the real-time monitoring personnel and is simultaneously distributed to the Drilling Fluids Engineer on the rig, the Company Representative (Company Man), office-based Fluids Engineers, and Subject Matter Experts (SMEs). This automated workflow effectively bridges the field and office environments, unlocking the full potential of centralized expertise. By enabling rapid response and collaborative problem-solving, the system supports timely recommendations, corrective actions, and fluid treatment instructions to ensure continued operational efficiency and wellbore integrity.

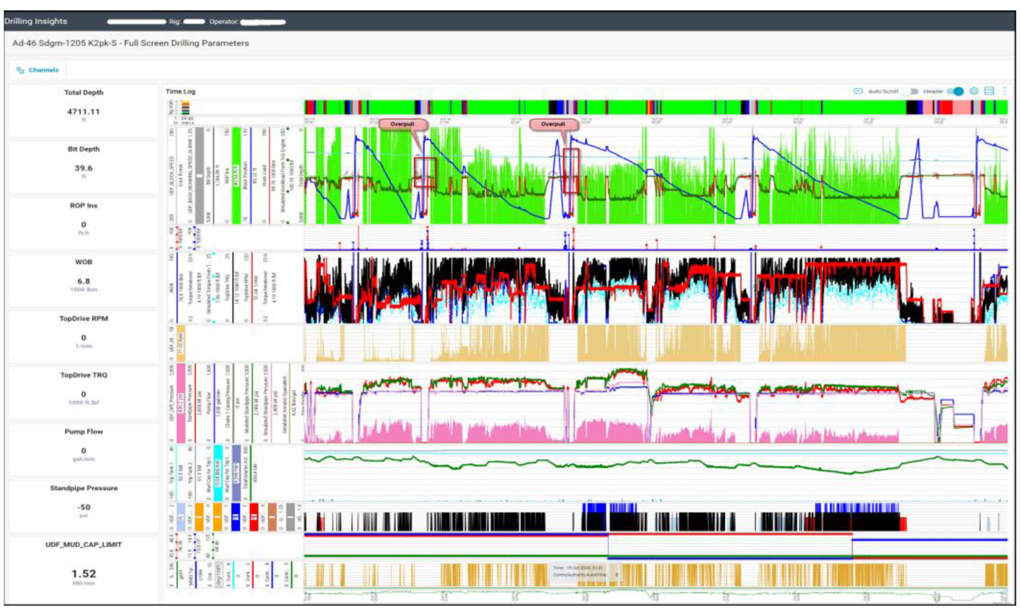

Figure 7—RheoProfiler real time data is Integrated to the main dashboard drilling visualization.

Results & Conclusions

The RheoProfiler system was systematically implemented across 11 Gas rigs, covering over 750,000 feet of drilling footage in 48 wells with 300 sections drilled within a single year. This extensive deployment addressed critical operational challenges related to drilling fluids, including well control, stuck pipe, fluid losses, high-pressure high-temperature (HPHT) conditions, and difficulties in reaching casing depth. The following key achievements were documented:

Prevention of 8 high-density differential sticking incidents through real-time adjustments enabled by accurate and timely fluid property data.

Mitigation of 7 well control events, allowing for safer and more efficient drilling operations.

Maintenance of optimal fluid behavior in HPHT environments, managing pressures up to 12,000 psi and temperatures of 150°C without compromising fluid stability or operational safety.

Elimination of aggressive mud dilutions or remedial treatments, achieving real-time Equivalent Circulating Density (ECD) control through precise density and rheology management, crucial during liner running in high-risk formations to avoid losses.

Improved safety outcomes by reducing manual measurement frequency, which freed engineers to focus on decision-making and minimized exposure to hazards, cumulatively saving 60 man-hours per month.

Enhanced overall drilling performance, with a significant reduction in operational hours across all wells drilled using the RheoProfiler system.

Accelerated Mud Cap management during loss scenarios, by substantially reducing fluid testing time and enabling rapid response.

Enhanced Data Frequency and Resolution

RheoProfiler increased test frequency from the conventional 2–3 manual tests per day to up to 6 automated tests daily, delivering unprecedented data resolution and enabling early detection of fluid anomalies.

Improved Accuracy and Consistency

The automation eliminated variability inherent in manual methods. Initial cross-validation showed deviations within ±2% for density and ±5% for rheology, confirming measurement accuracy under operational conditions.

Real-Time Data Availability

Data latency dropped from 24 hours to under 2 minutes, facilitating prompt operational responses that contributed to an estimated 17% reduction in fluids-related Non-Productive Time (NPT).

Operational Efficiency

Manual testing time per day decreased from approximately 90 minutes to 15 minutes, optimizing labor allocation and reducing errors associated with manual data entry.

Collaboration and Remote Support

Remote experts engaged proactively in over 45 interventions, providing guidance that directly impacted drilling efficiency and safety.

These results demonstrate RheoProfiler robust capability to provide reliable, real-time data under challenging drilling conditions, setting a new benchmark for fluid monitoring and control in Ghawar gas field operations and beyond.

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